

🌟 Graph Theory & Combinatorics Solutions: 2024

🌟 Q1 (i) Suppose G is a non-directed graph with 12 edges. If 6 vertices are each of degree 3 and the rest are degree less than 3, what is the minimum number of vertices in G ?

🌟 **Answer:**

◆ **Given:**

- Number of edges: $|E| = 12$
- 6 vertices each have a degree of 3.

💡 **Apply Handshaking Lemma:** According to the Handshaking Lemma: $\sum \deg(v) = 2|E|$

Therefore: $\sum \deg(v) = 2(12) = 24$

🔥 **Degree Contribution of Given 6 Vertices:** Each of the 6 vertices has a degree of 3. So the total degree contributed is: $6 \times 3 = 18$

🔍 **Remaining Degree Sum:** Remaining degree sum: $24 - 18 = 6$

Now:

- All remaining vertices must have a degree of less than 3.
- To minimize the number of vertices, we must maximize their degrees.

The maximum possible degree less than 3 is: **2**.

💡 **Minimum Remaining Vertices:** To obtain a degree sum of 6 using vertices of degree 2: $\frac{6}{2} = 3$

So the minimum additional vertices required is 3.

🌟 **Total Minimum Vertices:** $6 + 3 = 9$

✅ **Final Answer:** 🌟 Minimum number of vertices = **9**

🌟 Q1 (ii) Find the circuit matrix of the following graph G .

🌟 **Answer:**

◆ **Circuit Matrix:** A circuit matrix is defined as:

- rows represent circuits.
- columns represent edges.
- entry = 1 if the edge belongs to the circuit.
- otherwise, the entry is 0.

🔍 **Edges of Graph:** From the figure:

Edge	Connection
a	$V_1 \rightarrow V_2$
b	$V_3 \rightarrow V_2$
c	$V_4 \rightarrow V_3$
d	$V_1 \rightarrow V_4$
e	$V_2 \rightarrow V_4$
f	$V_2 \rightarrow V_3$
g	$V_1 \rightarrow V_5$

🔥 Identify Circuits:

Circuit 1: $V_1 \rightarrow V_2 \rightarrow V_4 \rightarrow V_1$ Uses: a, e, d

Circuit 2: $V_2 \rightarrow V_3 \rightarrow V_2$ Uses: b, f

Circuit 3: $V_1 \rightarrow V_2 \rightarrow V_3 \rightarrow V_4 \rightarrow V_1$ Uses: a, b, c, d

☀️ **Circuit Matrix:** Using the column order: a, b, c, d, e, f, g

$$B = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

✅ **Final Answer:**
$$B = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

☀️ Q1 (iii) What are the degrees of vertices in a graph G with no adjacent edges?

🌟 Answer:

💡 **Meaning of “No Adjacent Edges”:** No two edges share a common vertex. Such a graph is formally called a matching graph.

💡 **Degree of Vertices:** If no two edges are adjacent:

- a vertex can belong to at most one single edge.

Hence:

- the degree can only be 0 or 1.

☀️ Possibilities:

Vertex Type	Degree
Isolated vertex	0
Vertex incident to one edge	1

✓ **Final Answer:** In a graph with no adjacent edges, every vertex has a degree of either **0** or **1**.

☀️ Q2 (i) In an electric network consisting of 300 nodes and 500 cables connecting these nodes, find the minimum number of cables that can be removed to reduce the network to a spanning tree.

✨ **Answer:**

◆ **Spanning Tree Property:** A spanning tree with n vertices contains exactly: $n - 1$ edges.

🔍 **Given:**

- Nodes: $n = 300$
- Existing cables: $e = 500$

🔥 **Edges Required in Spanning Tree:** $300 - 1 = 299$

🔥 **Cables to Remove:** $500 - 299 = 201$

✓ **Final Answer:** ☀️ Minimum cables removed = **201**

☀️ Q2 (ii) Prove that "Every connected graph has a spanning tree."

✨ **Answer:**

◆ **Spanning Tree:** A spanning tree:

- contains all vertices.
- is connected.
- has no cycles.

💡 **Proof:** Suppose G is a connected graph.

If the graph has no cycles:

- then the graph itself is already a tree, and it serves as its own spanning tree.

If the graph contains cycles:

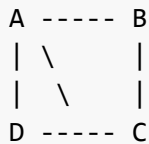
- remove one edge from a cycle.
- Removing one edge from a cycle does not disconnect the graph.

Repeat this exact process until all cycles are systematically removed. The resulting graph:

- remains connected.
- contains all vertices.
- has no cycles.

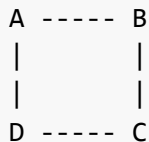
Hence, it perfectly becomes a spanning tree.

🌈 **Example:**



This graph contains a cycle. Let's remove edge AC .

Result:



Now it is:

- connected.
- contains no cycles.
- includes all vertices.

Hence, a spanning tree is successfully obtained.

✓ **Final Conclusion:** Every connected graph always has at least one spanning tree.

☀️ Q2 (iii) A rooted tree has n vertices and k pendant vertices. Find the number of vertices having degree 3.

✨ **Answer:**

◆ **Definitions:**

- Pendant vertex: degree = 1.
- Suppose the number of vertices having a degree of 3 is x .

💡 **Tree Property:** For any tree, the sum of degrees is: $\sum \deg(v) = 2(n-1)$

🔥 **Assume Remaining Vertices Have Degree 2:** Vertices breakdown:

- k pendant vertices of degree 1.
- x vertices of degree 3.
- remaining vertices of degree 2.

The number of remaining vertices is: $n - k - x$

🔥 **Degree Sum Equation:** $k(1) + 3(x) + 2(n - k - x) = 2(n - 1)$

Expand the equation: $k + 3x + 2n - 2k - 2x = 2n - 2$
 $x - k + 2n = 2n - 2$
 $x - k = -2$
 $x = k - 2$

✔ **Final Answer:** 🌟 Number of degree 3 vertices: $x = k - 2$

🌟 Q2 (iv) Using Kruskal’s Algorithm find the Minimal Spanning Tree of the following graph.

🌟 **Answer:**

♦ **Kruskal’s Algorithm:** Kruskal’s Algorithm constructs a Minimum Spanning Tree (MST) by:

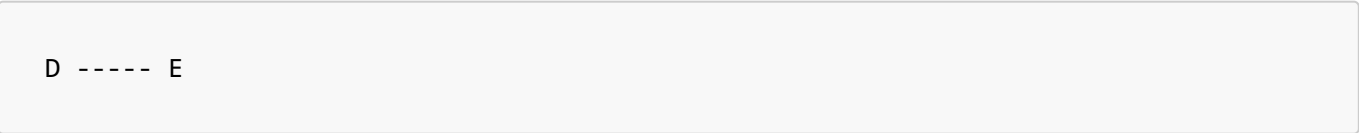
- Selecting edges in increasing order of weights.
- Avoiding cycles.
- Continuing until all vertices are connected.

🔍 **Step 1: List All Edges with Weights:** From the figure:

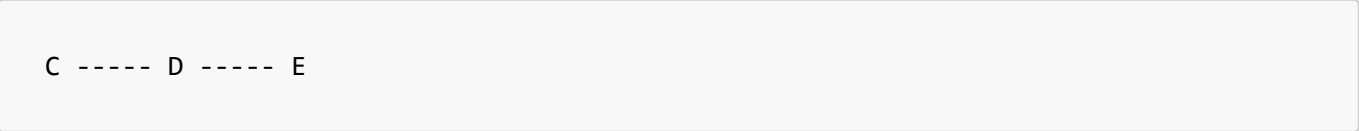
Edge	Weight
D–E	2
C–D	3
F–E	3
B–C	4
F–D	4
A–B	5
A–F	6
F–C	6
B–D	7
B–F	7
B–E	8

🔥 **Step 2: Select Smallest Edges:**

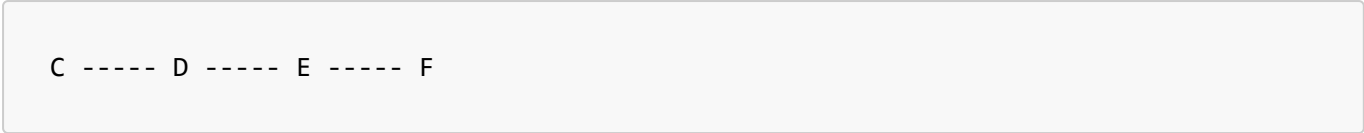
✔ Select $D-E = 2$. No cycle formed. Current MST:



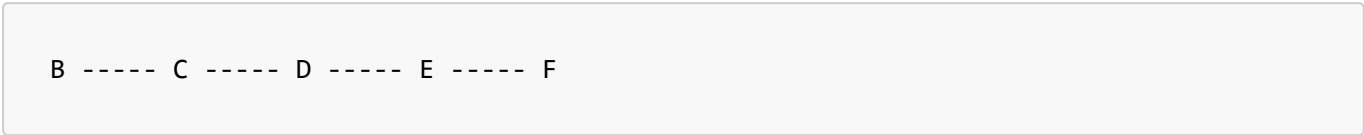
✔ Select $C-D = 3$. No cycle formed.



✔ Select $F-E = 3$. No cycle formed.



✔ Select $B-C = 4\$$. No cycle formed.



✗ Edge $F-D = 4\$$. This forms a cycle ($F-E-D-F\$$). Hence, reject it.

✔ Select $A-B = 5\$$. No cycle formed. Now all vertices are fully connected!

🌟 **Minimal Spanning Tree Edges:**

Edge	Weight
D–E	2
C–D	3
F–E	3
B–C	4
A–B	5

🔥 **Total Weight:** $2 + 3 + 3 + 4 + 5 = 17\$$

✔ **Final Answer:** The Minimum Spanning Tree contains edges: $\{(D,E), (C,D), (F,E), (B,C), (A,B)\}$. Total minimum cost: 🌟 **17**

🌟 Q3 (i-a) Identify all the cut edges in the graph, if any and explain your process.

🌟 **Answer:**

💠 **Cut Edge (Bridge):** A cut edge is an edge whose removal disconnects the graph.

🔍 **Given Graph Details:** Vertices: A, B, C, D, E, F Edges: $\{A,B\}, \{A,C\}, \{A,D\}, \{B,D\}, \{B,E\}, \{C,F\}, \{D,E\}, \{E,F\}$

💡 **Check Each Edge:** We systematically examine whether removing an edge disconnects the graph.

🔥 **Edge $C-F$:** If removed, C becomes totally disconnected from the F -side of the graph. No alternative path exists. Hence, $C-F$ is a cut edge! ✔

🔥 **Other Edges:** All remaining edges belong to cycles. Removing them means the graph still remains connected via alternative routes. Hence they are not cut edges.

✔ **Final Answer:** 🌟 Cut Edge: $\{C,F\}$

☀️ Q3 (i-b) Determine if there are any cut vertices, and if so, identify them.

✨ **Answer:**

♦ **Cut Vertex:** A cut vertex is a vertex that disconnects the graph when it is removed.

🔍 **Check Vertices:**

🔥 **Vertex F :** Removing F completely disconnects C from the rest of the graph. Hence, F is a cut vertex! ✅

🔥 **Vertex C :** Removing C does not disconnect the core graph (it only isolates itself, which is just making the graph smaller). Hence, it is not a cut vertex.

🔥 **Vertex E :** Removing E leaves the graph still connected through other paths (like B and D). Hence, it is not a cut vertex.

✅ **Final Answer:** ☀️ Cut Vertex: F

☀️ Q3 (i-c) Determine if there are any minimal cut set that would disconnect node A from node F .

✨ **Answer:**

♦ **Minimal Cut Set:** A minimal cut set is the smallest set of edges whose removal disconnects two specified vertices.

🔍 **Paths from A to F :** Possible direct/indirect paths:

- $A \rightarrow C \rightarrow F$
- $A \rightarrow B \rightarrow E \rightarrow F$
- $A \rightarrow D \rightarrow E \rightarrow F$

🔥 **Disconnecting A and F :** Removing only one edge is insufficient because alternative paths still exist.

✅ **Minimal Cut Set:** One minimal cut set is: (A,C) and (E,F) . After removing these edges, absolutely no path remains between A and F .

🎯 **Final Answer:** A minimal cut set disconnecting A and F is: ☀️ $\{(A,C), (E,F)\}$

☀️ Q3 (ii) Justify using a suitable example: "The vertex connectivity of any graph G can never exceed the edge connectivity of G ."

✨ **Answer:**

♦ **Definitions:**

- **Vertex Connectivity $K(G)$** = minimum number of vertices whose removal disconnects the graph.
- **Edge Connectivity $\lambda(G)$** = minimum number of edges whose removal disconnects the graph.

 **Important Theorem:** $K(G) \leq \lambda(G)$

 **Example: Complete Graph K_4**



 **Vertex Connectivity:** We need to remove at least 3 vertices to disconnect it. Hence: $K(G) = 3$

 **Edge Connectivity:** We need to remove at least 3 edges to disconnect it. Hence: $\lambda(G) = 3$

 **Verification:** $K(G) = \lambda(G)$ Thus, the theorem is perfectly satisfied.

 **Final Conclusion:** Vertex connectivity can never exceed edge connectivity.

 Q3 (iii) Compute the union of graphs G and H .

 **Answer:**

 **Given Graphs:**

Graph G : Vertices: $V_1 = \{A, B, C, D, E, F\}$ Edges: $E_1 = \{(A,B), (A,C), (B,C), (B,D), (C,E), (D,F), (E,F)\}$

Graph H : Vertices: $V_2 = \{B, D, E, F, G\}$ Edges: $E_2 = \{(B,D), (D,E), (D,F), (E,F), (E,G)\}$

 **Union of Graphs:**

Vertex Set: $V(G \cup H) = V_1 \cup V_2 = \{A, B, C, D, E, F, G\}$

Edge Set: $E(G \cup H) = E_1 \cup E_2 = \{(A,B), (A,C), (B,C), (B,D), (C,E), (D,F), (E,F), (D,E), (E,G)\}$

 **Final Answer:**

- **Vertices:** $\{A, B, C, D, E, F, G\}$
- **Edges:** $\{(A,B), (A,C), (B,C), (B,D), (C,E), (D,F), (E,F), (D,E), (E,G)\}$

 Q4 (i) "A graph is 2-colorable if it is bipartite and every cycle has an even length." Justify this with a suitable example.

 **Answer:**

 **2-Colorable Graph:** A graph is said to be 2-colorable if:

- only two colors are enough to color all vertices. 

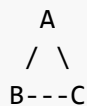
- no two adjacent vertices receive the same color.

♦ **Bipartite Graph:** A graph is bipartite if:

- vertices can be divided into two disjoint sets.
- edges connect vertices only from different sets.

💡 **Fundamental Theorem:** ✅ A graph is bipartite if and only if every cycle has an even length. This theorem is extremely important in graph coloring.

🔥 **Why Odd Cycles Fail:** Suppose we try to color a triangle:

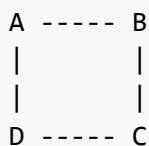


Cycle length is 3, which is odd. Try using two colors:

- A = Red
- B = Blue
- C must differ from B $\rightarrow$ Red

But now A and C both become Red, and they are adjacent! ❌ Hence, odd cycles are fundamentally not 2-colorable.

🌈 **Suitable Example of 2-Colorable Graph:** Consider a square graph:



This is a cycle C_4 . Length is 4, which is EVEN ✅

🎨 **Proper Coloring:** Assign colors:

Vertex	Color
A	Red
B	Blue
C	Red
D	Blue

No adjacent vertices have the same color. Thus, the graph is bipartite and perfectly 2-colorable.

🎯 **Final Conclusion:** A graph is 2-colorable exactly when it is bipartite and every cycle has an even length.

☀️ Q4 (ii) Find the chromatic number of the graph given below.

🌟 Answer:

♦ **Chromatic Number:** The chromatic number of a graph is $\chi(G)$, which is the minimum number of colors needed to color the graph such that adjacent vertices receive different colors.

🔍 **Observation of the Given Graph:** The graph contains:

- several interconnected cycles.
- odd cycles.
- intersecting regions.

Hence, at least 3 colors are strictly required.

🎨 **Coloring Process:** Use colors: ● Red, ● Blue, ● Green. Assign colors carefully to ensure no adjacent vertices share the same color.

Possible assignment:

Vertex Group	Color
A, C, H	●
B, D, K	●
E, F, J, L, M, G	●

This satisfies proper coloring perfectly.

❌ **Why 2 Colors are Impossible:** The graph contains odd cycles. Odd cycles are not bipartite. Therefore, the graph absolutely cannot be colored using only 2 colors.

✅ **Final Answer:** $\chi(G) = 3$ ☀️ Chromatic Number = 3

☀️ Q4 (iii-a) Find the maximum matching in the bipartite graph.

🌟 Answer:

♦ **Given Sets:**

- Set U: $\{A, B, C, D\}$
- Set V: $\{1, 2, 3, 4\}$

♦ **Given Edges:** $(A,1), (A,2), (B,2), (B,3), (C,3), (C,4), (D,4)$

🔥 **Maximum Matching:** A matching is maximum if:

- the largest possible number of vertices are matched.
- no two edges share a common vertex.

🌈 **One Maximum Matching:**

- $(A,1)$
- $(B,2)$
- $(C,3)$
- $(D,4)$

 **Number of Matching Edges:** There are **4** matching edges. Each edge matches one vertex from U and one vertex from V. Hence, all 8 vertices become matched!

✓ **Final Answer:** Maximum matching: $\{(A,1), (B,2), (C,3), (D,4)\}$ Number of matching edges:  **4**

 **Q4 (iii-b)** Find a maximal matching in the graph and list the matching edges.

 **Answer:**

♦ **Maximal Matching:** A maximal matching means it cannot be extended further; adding any extra edge strictly violates the matching condition.

 **One Maximal Matching:**

- $(A,2)$
- $(B,3)$
- $(C,4)$

 **Why Maximal?** Now, no more edges can be added because every remaining edge shares a vertex with the currently chosen edges. Hence, this matching is definitely maximal.

✓ **Final Answer:** One maximal matching is:  $\{(A,2), (B,3), (C,4)\}$

 **Q4 (iii-c)** Determine if the graph has a perfect matching.

 **Answer:**

♦ **Perfect Matching:** A perfect matching means every single vertex of both sets is matched exactly once.

 **Check Matching:** The matching $\{(A,1), (B,2), (C,3), (D,4)\}$ covers:

- all vertices in Set U.
- all vertices in Set V.

Hence, every vertex is matched exactly once.

✓ **Therefore:** The graph absolutely HAS a perfect matching.

 **Perfect Matching:** $\{(A,1), (B,2), (C,3), (D,4)\}$

 **Final Conclusion:**

Type	Exists?
Maximum Matching	✓ Yes

Type	Exists?
Maximal Matching	✓ Yes
Perfect Matching	✓ Yes

☀️ Q5 (i) What is the minimum number of students required in a Graph Theory class to be sure that at least six will receive the same grade, if there are five possible grades A, B, C, D and F?

✨ **Answer:**

♦ **Apply Pigeonhole Principle** 📌 Possible grades: A, B, C, D, F. Total grade categories: **5**

🔥 **Worst Case Scenario:** To avoid having 6 students in the same grade, we place at most 5 students in each grade. Thus: $5 \times 5 = 25$

At this stage, no grade has 6 students yet. One more student naturally guarantees that some grade gets 6 students.

✓ **Final Answer:** $25 + 1 = 26$ ☀️ Minimum students required = **26**

☀️ Q5 (ii) Suppose a club has 25 members. How many ways are there to choose a president, vice-president, secretary and treasurer where no person can hold more than one post?

✨ **Answer:**

♦ **Important Observation:** Posts are different:

- President
- Vice-President
- Secretary
- Treasurer

Hence, order matters. So, permutations are correctly used here.

🔥 **Selection Process:**

- President: **25** choices
- Vice-President: **24** choices
- Secretary: **23** choices
- Treasurer: **22** choices

🔥 **Total Ways:** $25 \times 24 \times 23 \times 22 = 303600$

✓ **Final Answer:** ☀️ Total ways = **303,600**

☀️ Q5 (iii) Seven women and nine men are on the faculty in the IT department at a University. How many ways are there to select a

committee of five members if at least one woman and at least one man must be on the committee?

✨ **Answer:**

◆ **Total Faculty Members:**

- Women: **7**
- Men: **9**
- Total: **16**
- Committee size: **5**

🔥 **Method:** We strategically use the logic: $\text{Required ways} = \text{Total ways} - \text{All men} - \text{All women}$

🔥 **Total Committees:** $\binom{16}{5} = 4368$

🔥 **All-Men Committees:** $\binom{9}{5} = 126$

🔥 **All-Women Committees:** $\binom{7}{5} = 21$

🔥 **Required Committees:** $4368 - 126 - 21 = 4221$

✅ **Final Answer:** ☀️ Number of committees = **4221**

☀️ Q6 (i) Find the generating function of the sequence in closed form:
 $\langle 4, 4, 4, 4, 4, \dots \rangle$

✨ **Answer:**

◆ **Given Sequence:** The sequence is: $4, 4, 4, 4, \dots$ This is a constant sequence where every single term equals 4.

💡 **Recall Basic Generating Function:** We know the standard identity: $1 + x + x^2 + x^3 + \dots = \frac{1}{1-x}$

🔥 **Multiply by 4:** Since every term is 4, we have: $4 + 4x + 4x^2 + 4x^3 + \dots$

Factor out the 4: $= 4(1 + x + x^2 + x^3 + \dots)$

Using the geometric series formula: $= 4 \left(\frac{1}{1-x} \right)$

✅ **Final Answer:** $\frac{4}{1-x}$

☀️ **Explanation:** This generating function effortlessly expands as: $\frac{4}{1-x} = 4 + 4x + 4x^2 + 4x^3 + \dots$ which flawlessly generates the sequence: $\langle 4, 4, 4, 4, \dots \rangle$.

☀️ Q6 (ii) Using generating functions, find a_n in terms of n for the recurrence relation: $a_0 = 1$, $a_1 = 2$ and $a_{n+2} = 5a_{n+1} - 4a_n$, $n \geq 0$

🌟 Answer:

◆ **Step 1: Assume Generating Function** Let: $A(x) = a_0 + a_1x + a_2x^2 + \dots$

We are given the linear recurrence relation: $a_{n+2} = 5a_{n+1} - 4a_n$

🔥 **Step 2: Form Characteristic Equation** Convert the recurrence relation directly into its characteristic equation: $r^2 - 5r + 4 = 0$

Factorize: $(r - 4)(r - 1) = 0$

Roots: $r = 4, \quad r = 1$

🔥 **Step 3: General Solution** Since the roots are perfectly distinct: $a_n = A(4^n) + B(1^n)$ $a_n = A(4^n) + B$

🔥 **Step 4: Use Initial Conditions** Using $a_0 = 1$: $A + B = 1$

Using $a_1 = 2$: $4A + B = 2$

Subtract the equations: $(4A + B) - (A + B) = 2 - 1$ $3A = 1$ $A = \frac{1}{3}$

Now solve for B : $B = 1 - \frac{1}{3}$ $B = \frac{2}{3}$

🔥 **Step 5: Final Formula** Substitute the values back in: $a_n = \frac{1}{3}(4^n) + \frac{2}{3}$

✅ **Final Answer:** $a_n = \frac{4^n + 2}{3}$

🌈 **Verification:** For $n=0$: $a_0 = \frac{1+2}{3} = 1$ (Correct ✅)

For $n=1$: $a_1 = \frac{4+2}{3} = 2$ (Correct ✅)

🎯 **Final Conclusion:** The required explicit formula is definitively: 🌟 $a_n = \frac{4^n + 2}{3}$